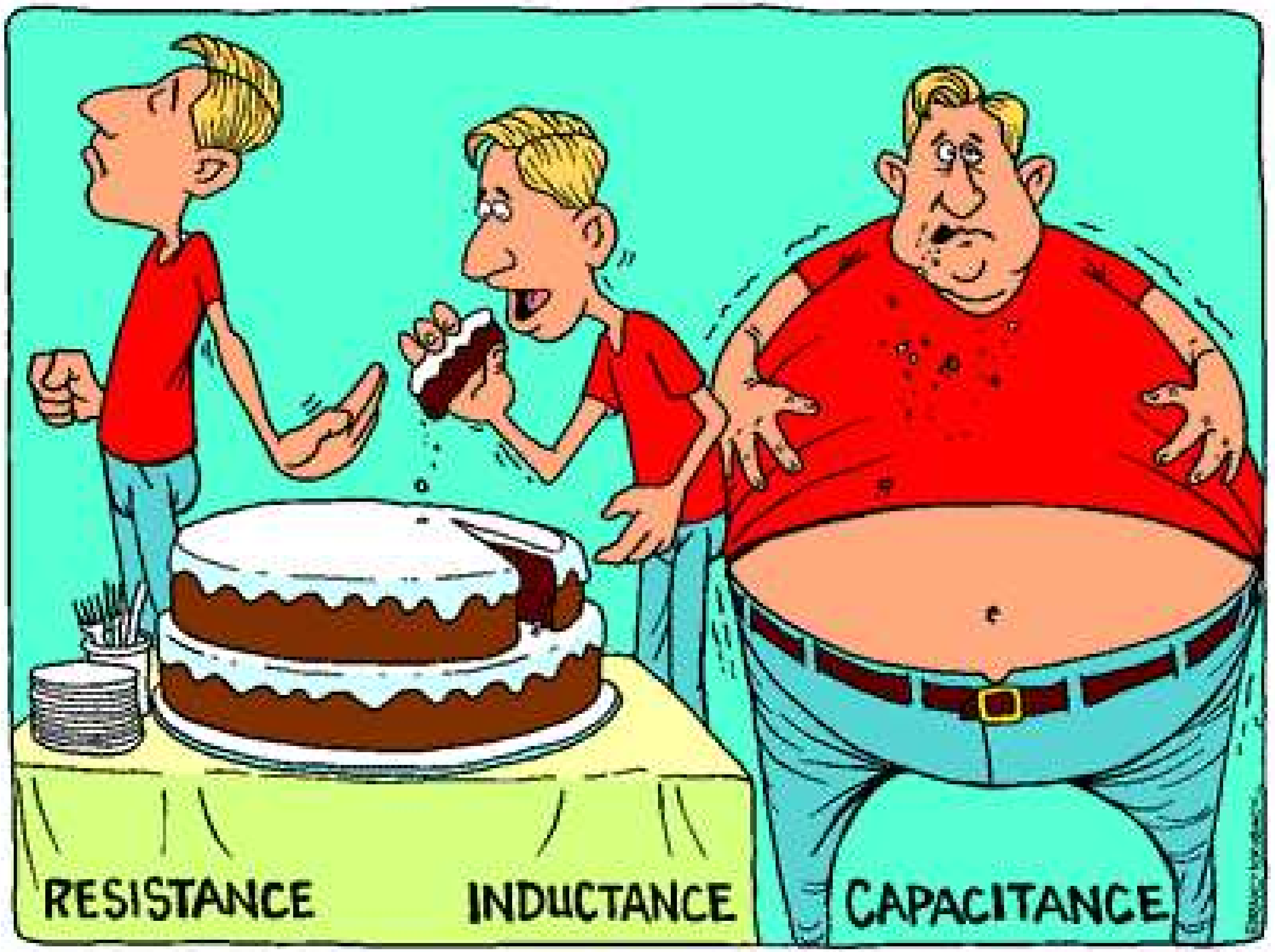
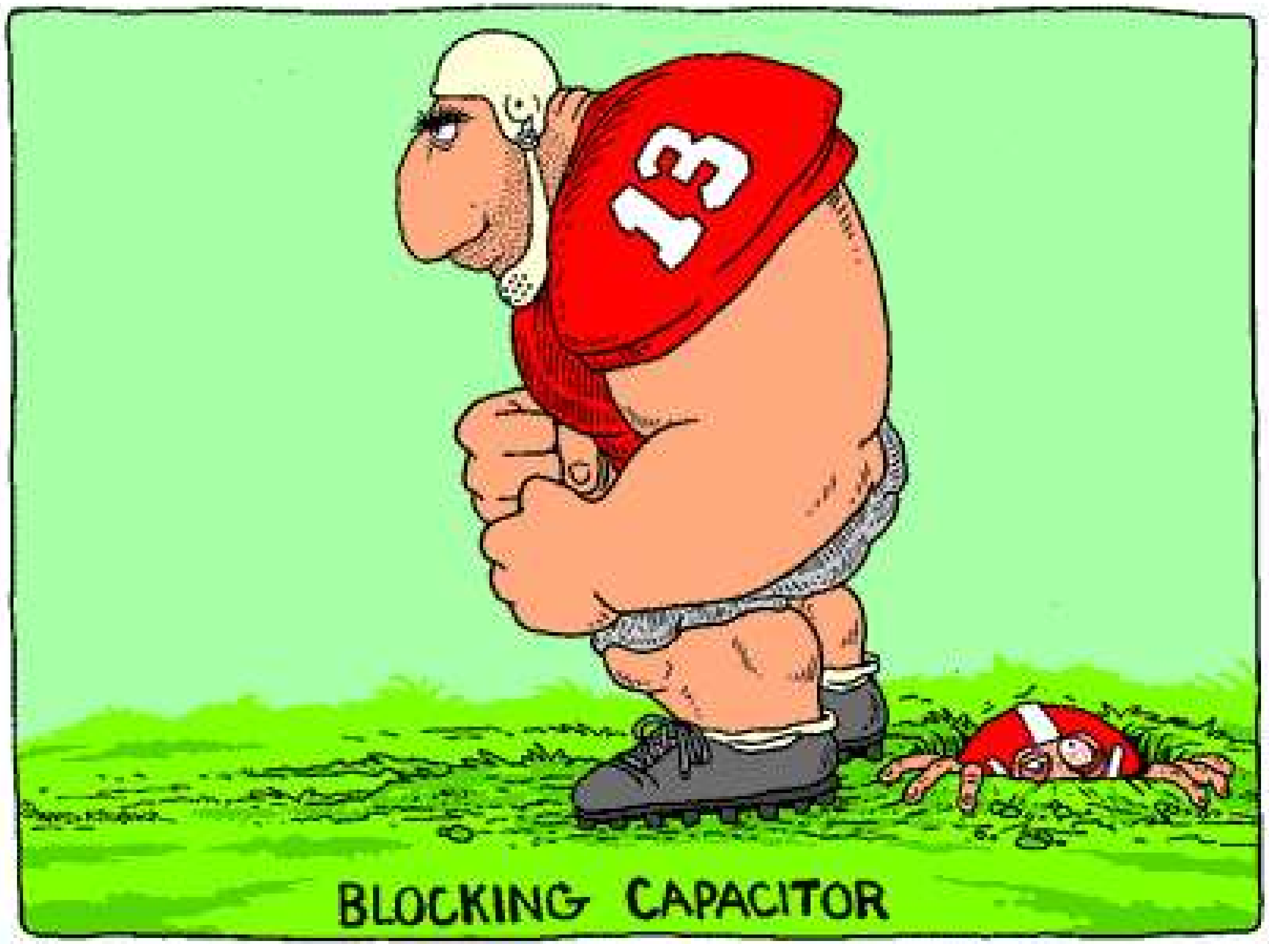
P-N junction - Capacitance

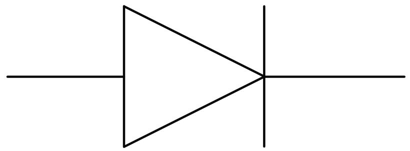


P-N junction - Capacitance

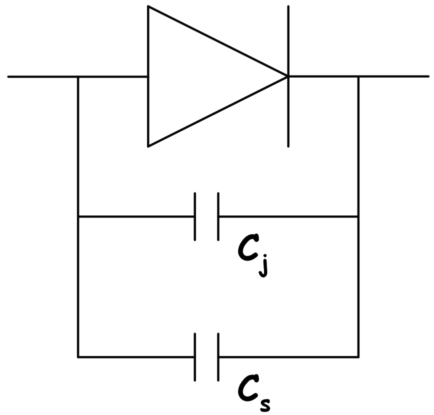


P-N junction - Capacitance

Two kinds of capacitances are associated with pn junctions. These are junction capacitance (or transition capacitance) and charge storage capacitance (or diffusion capacitance). These are modeled by two capacitors, C_j and C_s respectively, connected parallel to the pn junction.



Ideal pn junction



Practical pn junction

P-N junction - Capacitance

Junction capacitance has physical characteristics similar to that of a parallel-plate capacitor and the expression that defines this capacitance is identical to that of a parallel-plate capacitor.

The storage capacitance also possesses the charge storing property of a capacitor but has no other characteristics similar to that of a parallel-plate capacitor.

Capacitance is present in any device if a small change in the applied voltage **dV** across it results in a change in some charge **dQ** stored in it. By definition, the capacitance ***C*** is given by

$$C = \frac{dQ}{dV} \qquad (49)$$

P-N junction - Capacitance

As the voltage **V** across a pn junction varies, there are 2 stored charges in it, $\mathbf{Q_j}$ and $\mathbf{Q_s}$, that will be affected. Hence, there are 2 associated capacitances.

$$C_j = dQ_j/dV,\ C_s = dQ_s/dV$$

From circuit point of view, the voltage across the pn junction cannot change instantaneously due to the presence of C_j and C_s. This implies that there will be delay in circuits involving pn junctions as the capacitors C_j and C_s will need time to be charged up or discharged.

Physically, this delay is related to the fact that <u>a finite time is required for current to flow and charges to be transported to or away from the pn junction, so that the stored charge $\mathbf{Q_j}$ and $\mathbf{Q_s}$ can be built-up or reduced to their new value upon changing **V**.</u>

P-N junction – Junction Capacitance

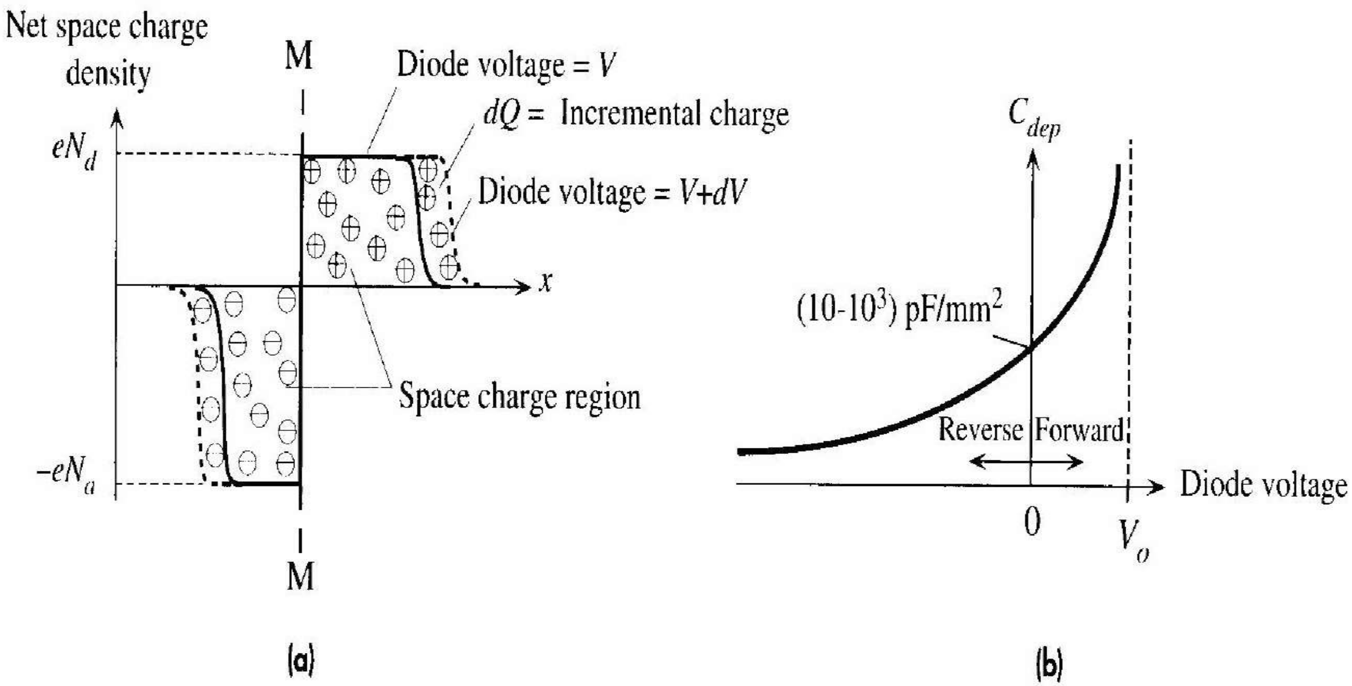


Figure 0.11 The depletion region behaves like a capacitor.
(a) The charge in the depletion region depends on the applied voltage just as in a capacitor.
(b) The incremental capacitance of the depletion region increases with forward bias and decreases with reverse bias. Its value is typically in the range of picofarads per mm^2 of device area.

P-N junction – Junction Capacitance

Upon changing the bias **V** across a pn junction, the depletion width **W** and hence the amount of space charge in the depletion region will be affected.

The space charge $\mathbf{Q_j}$ in the depletion region **W** is given by

$$Q_j = |Q| = q\,A\,x_{po}\,N_a = q\,A\,x_{no}\,N_d \qquad (50)$$

Under FB, **W** decreases and hence $\mathbf{Q_j}$ decreases. Under RB, **W** increases and hence $\mathbf{Q_j}$ increases. Thus,there is a capacitive effect associated with this change of charge $\mathbf{Q_j}$. Unlike the parallel plate capacitor, $\mathbf{Q_j}$ is not proportional to **V**, viz., non-linear, and hence eqn. (50) must be used to determine $\boldsymbol{C_j}$.

P-N junction – Junction Capacitance

With applied bias **V** (+ve when FB and –ve when RB)

$$W = \left[\frac{2\,\varepsilon\,(V_o - V)}{q} \left(\frac{N_a + N_d}{N_a N_d} \right) \right]^{1/2} \qquad (51)$$

Now, recall the expressions for x_{no} and x_{po}

$x_{no} = WN_a / (N_a + N_d)$ and $x_{po} = WN_d / (N_a + N_d)$

Thus, the space charge $\mathbf{Q_j}$ can now be written as

$$Q_j = q\,A\,\frac{N_a N_d}{N_a + N_d}\,W$$

$$= A \left[2\,q\,\varepsilon\,(V_o - V) \frac{N_a N_d}{N_a + N_d} \right]^{1/2} \qquad (52)$$

P-N junction – Junction Capacitance

The junction capacitance C_j is thus given by

$$C_j = \left| \frac{dQ_j}{dV} \right| = \left| \frac{dQ_j}{d(V_o - V)} \right| = \frac{A}{2} \left[\frac{2q\varepsilon}{(V_o - V)} \frac{N_a N_d}{N_a + N_d} \right]^{1/2} \quad (53)$$

Note that C_j is voltage dependent (**$C_j \propto (V_o - V)^{-1/2}$**).
Rearranging eqn. (53) yields

$$C_j = \varepsilon A \left[\frac{q}{2\varepsilon(V_o - V)} \frac{N_a N_d}{N_a + N_d} \right]^{1/2} = \frac{\varepsilon A}{W} \quad (54)$$

P-N junction – Junction Capacitance

The expression for $\mathbf{C_j}$ is an exact analogy with that of a parallel plate capacitor, with the depletion width **W** corresponding to the plate separation of the capacitor.

Physically as we change the bias **V** across the pn junction, it takes time for the majority carriers to respond in order to expose a larger space charge region under RB, or a smaller space charge region under FB.

$\mathbf{C_j}$ is associated with this delay time for **W** to reach its steady state width.

In other words, **W** cannot change instantaneously in response to a sudden change in the bias.

P-N junction – Junction Capacitance

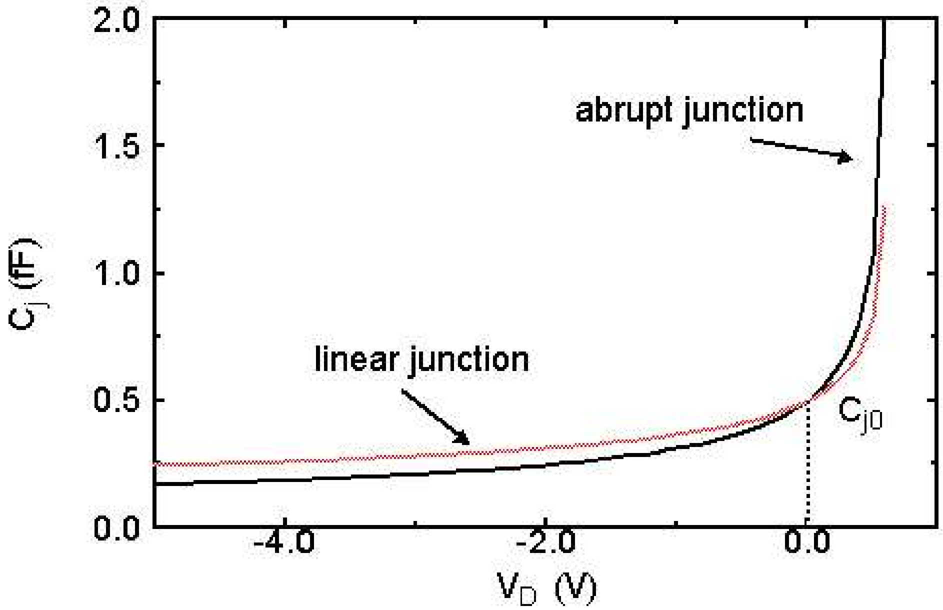
It is quite common to write the expression for C_j as a function of applied voltage. C_j is often written as

$$C_j(V) = C_{jo}\left[\frac{V_o}{V_o - V}\right]^{1/2}$$

Where C_{jo} is the zero-bias junction capacitance, given by

$$C_{jo} = \varepsilon A\left[\frac{q}{2\varepsilon}\frac{1}{V_o}\frac{N_a N_d}{N_a + N_d}\right]^{1/2}$$

Junction Capacitance



$$C_j(V) = C_{jo}\left[\frac{V_o}{V_o - V}\right]^m$$

m = 0.5: abrupt junction
m = 0.33: linear junction

P-N junction – Junction Capacitance - Example

The doping densities of an abrupt-junction silicon PN diode are $N_a = 10^{17}/cm^3$ and $N_d = 8\times10^{15}/cm^3$ and the area of the junction is $2\times10^{-5}\ cm^2$. Calculate the junction capacitance at (a) zero bias, (b) RB of 6V and (c) FB of 0.7 V. Assume $n_i = 10^{10}\ cm^{-3}$.

Solution: First, we have

$$C_j = \varepsilon A \left[\frac{q}{2\varepsilon (V_o - V)} \frac{N_a N_d}{N_a + N_d} \right]^{1/2} = \frac{\varepsilon A}{W}$$

Also, the equilibrium contact potential V_o is

$$V_o = \frac{kT}{q} \ln \left(\frac{N_a N_d}{n_i^2} \right)$$

P-N junction – Junction Capacitance - Example

Calculate V_o as

V_o = 0.0259 ln[$(10^{17}\times 8 \times 10^{15})/(10^{10})^2$] = 0.77 V

(a) At zero bias

$$C_j = 11.8 \times 8.85 \times 10^{-14} \times 2 \times 10^{-5} \times \left[\frac{1.6 \times 10^{-19}}{2 \times 11.8 \times 8.85 \times 10^{-14} (0.77 - 0)} \frac{10^{17}\ 8 \times 10^{15}}{10^{17} + 8 \times 10^{15}} \right]^{1/2}$$

C_j = 0.56 pF

P-N junction – Junction Capacitance - Example

(b) With RB of 6 V,

$$C_j = 11.8 \times 8.85 \times 10^{-14} \times 2 \times 10^{-5} \times \left[\frac{1.6 \times 10^{-19}}{2 \times 11.8 \times 8.85 \times 10^{-14} (0.77 + 6)} \frac{10^{17} \; 8 \times 10^{15}}{10^{17} + 8 \times 10^{15}} \right]^{1/2}$$

C_j = 0.188 pF

P-N junction – Junction Capacitance - Example

(b) With FB of 0.7 V,

$$C_j = 11.8 \times 8.85 \times 10^{-14} \times 2 \times 10^{-5} \times$$

$$\left[\frac{1.6 \times 10^{-19}}{2 \times 11.8 \times 8.85 \times 10^{-14} (0.77 - 0.7)} \frac{10^{17} \; 8 \times 10^{15}}{10^{17} + 8 \times 10^{15}} \right]^{1/2}$$

$$C_j = 1.85 \text{ pF}$$

Note that junction capacitance in FB is much larger than that in RB as the width of the depletion region is much smaller in FB.

P-N junction – Storage Capacitance

The application of FB to a diode results in a reduction in the barrier height, a reduction in the depletion region width, and an injection of minority carriers across the depletion region into the opposite region, where they are stored as excess minority carriers. The density of the excess stored minority carriers increases with an increase in the FB.

Note that with changing bias **V**, there will be a change in the minority carrier concentrations at both sides of the junction, near the depletion region edges.

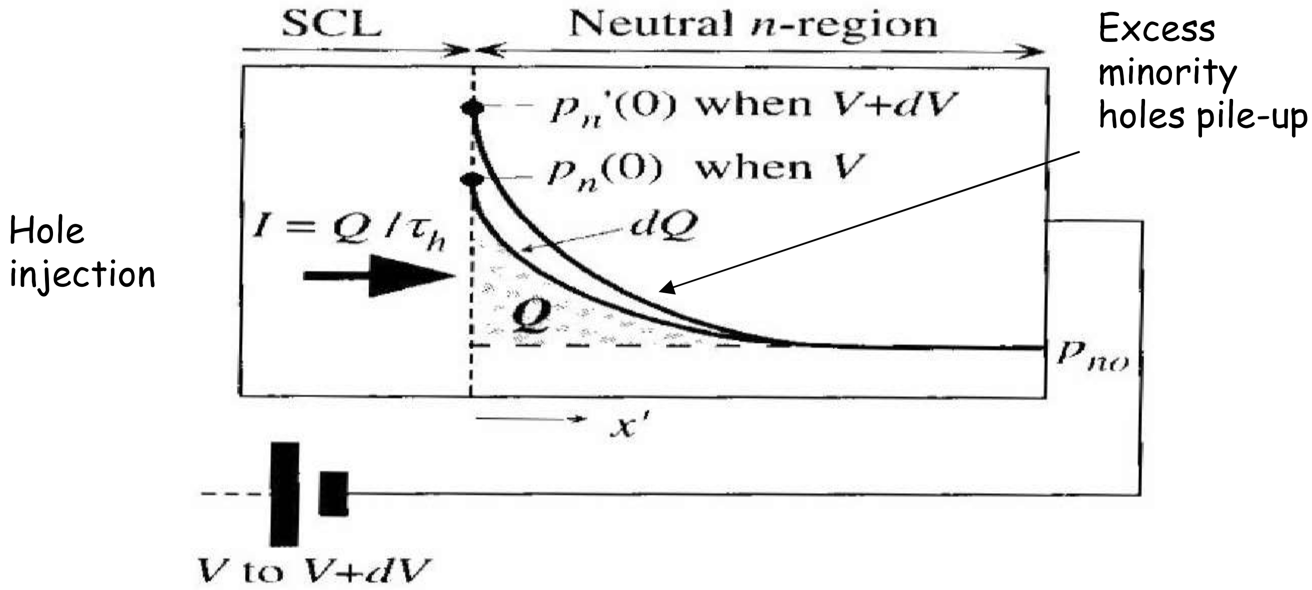
They become larger upon FB or smaller upon RB.

The increase in the minority carrier concentration does not take place instantaneously with a sudden change in FB.

A delay time is involved in the minority carrier concentration attaining a steady-state.

This delay is due to capacitive effect as a result of stored minority carrier charges in the neutral n and p regions. This is the origin of storage capacitance.

P-N junction – Storage Capacitance



Consider the injection of holes into the *n*-side during forward bias. Storage or diffusion capacitance arises because when the diode voltage increases from V to $V + dV$, more minority carriers are injected and more minority carrier charge is stored in the *n*-region.

P-N junction – Storage Capacitance

The <u>total excess stored charge for minority holes at n-side</u> $\mathbf{Q_p}$ is

$$Q_p = qA\int_0^\infty \delta p(x_n)\, dx_n = qA\,\Delta p_n \int_0^\infty e^{-x_n/L_p}\, dx_n$$

$$= qA\, L_p\, \Delta p_n \qquad (55)$$

The <u>total excess stored charge for minority electron at p-side</u> $\mathbf{Q_n}$ is

$$Q_n = -qA\int_0^\infty \delta n(x_p)\, dx_p = -qA\,\Delta n_p \int_0^\infty e^{-x_p/L_n}\, dx_p$$

$$= -qA\, L_n\, \Delta n_p \qquad (56)$$

P-N junction – Storage Capacitance

Here, $\delta \mathbf{p}(\mathbf{x}_n)$ and $\delta \mathbf{n}(\mathbf{x}_p)$ are the steady state profiles of the minority holes and electrons respectively, given by

$$\delta n(x_p) = \Delta n_p \, e^{-x_p/L_n} = n_{po}(e^{qV/kT} - 1)\, e^{-x_p/L_n}$$

$$\delta p(x_n) = \Delta p_n \, e^{-x_n/L_p} = p_{no}(e^{qV/kT} - 1)\, e^{-x_n/L_p}$$

Let $\mathbf{Q}_s$ denote the total excess charge stored associated with the <u>minority carriers</u>, viz., $\mathbf{Q}_s = \mathbf{Q}_p + \mathbf{Q}_n$.

Consider the case of a p^+n junction under FB. Recall that <u>the hole injection current will dominate</u>, viz., $\mathbf{I}_p(\mathbf{x}_p=0) \gg \mathbf{I}_n(\mathbf{x}_n=0)$. This transforms to $\Delta \mathbf{p}_n \gg \Delta \mathbf{n}_p$, and hence $\mathbf{Q}_p \gg \mathbf{Q}_n$.

P-N junction – Storage Capacitance

Hence,

$$C_s = dQ_s/dV \approx dQ_p/dV \qquad (57)$$

Recall

$$\Delta p_n = p_n(x_{no}) - p_{no} = p_{no}(e^{qV/kT} - 1)$$

Thus, the storage charge capacitance C_s associated with this change of storage charge is

$$C_s = \frac{d\,Q_p}{d\,V} = \frac{q^2}{kT} A\ L_p\ p_{no}\ e^{qV/KT} = \frac{q}{kT} Q_p \qquad (58)$$

P-N junction – Storage Capacitance

For $p^{+}n$ junction hole injection current will dominate and hence the total current **I** is given by

$$I \approx \frac{q\,A\,D_p}{L_p}\,\Delta\,p_n = \frac{q\,A\,L_p\,\Delta\,p_n}{\tau_p} = \frac{Q_p}{\tau_p} \qquad (59)$$

Thus, $\mathbf{Q_p} \approx \mathbf{I}\,\tau_p$.

Hence, C_s can also be expressed in terms of **I** and τ_p as

$$C_s = \frac{q}{kT}\,I\,\tau_p \qquad (60)$$

P-N junction – Storage Capacitance

It can be seen that $\boldsymbol{C_s}$ can be very large under FB due to the large **I**.

Physically, this can be understood because the storage charge depends exponentially on the bias voltage **V** under FB, meaning that there will be a large change in the amount of storage charge ($\mathbf{dQ_s}$) of the minority carriers with changing bias (**dV**), resulting in a large $\boldsymbol{C_s}$.

<u>The $\boldsymbol{C_s}$ will impose a serious limitation for a FB biased pn junction in high frequency applications</u>.

P-N junction – Storage Capacitance

The total capacitance across the pn junction is $C_T = C_j + C_s$.

<u>During FB, C_s is dominant</u> ($C_s \gg C_j$) due to the large change in the amount of storage charge of the minority carriers, hence $C_T \approx C_s$.

During RB, C_s is not critical ($C_s \ll C_j$) since there is only very small change in the amount of storage charge of the minority carriers. In comparison, <u>the junction capacitance C_j dominates under RB</u>, viz., $C_T \approx C_j$.

The presence of these capacitances will limit the speed of operation of any circuits using pn junction diodes.

P-N junction – Storage Capacitance - Example

The doping densities of an abrupt-junction silicon PN diode are $N_a = 10^{17}/cm^3$ and $N_d = 8 \times 10^{15}/cm^3$ and the area of the junction is 2×10^{-5} cm^2. The lifetime of holes in the n region is 0.1 μs. Given the diffusion constant for holes in N is 16 cm^2/s, calculate, at room temperature, the storage capacitance at (a) FB of 0.6 V and (b) 0.65 V. Assume $n_i = 1.5\times 10^{10}$ cm^{-3}.

Solution: Recall

$$C_S = \frac{q^2}{KT} A\ L_p\ p_{no}\ e^{qV/kT}$$

Equilibrium density of holes in the N region is

$p_{no} = n_i^2/N_d = 10^{20}/8 \times 10^{15} = 1.25 \times 10^4/cm^3$

Also, $L_p = (D_p\tau_p)^{1/2} = (16 \times 0.1\times10^{-6})^{1/2} = 1.26\times10^{-3}$ cm

P-N junction – Storage Capacitance - Example

(a) When the FB is 0.6 V, storage capacitance is

$$C_S = \frac{(1.6\times10^{-19})^2}{0.026\times1.6\times10^{-19}} 2\times10^{-5}\times 1.26\times10^{-3}\times1.25\times10^{4}\ e^{0.6/0.026}$$

C_s **= 20.4 pF**

(b) Similarly, at FB of 0.65 V

$$C_S = \frac{(1.6\times10^{-19})^2}{0.026\times1.6\times10^{-19}} 2\times10^{-5}\times 1.26\times10^{-3}\times1.25\times10^{4}\ e^{0.65/0.026}$$

C_s **= 139.6 pF**

Note the large value of C_s. It is in fact much larger than C_j in FB (compare with the previous example for C_j value. It was 1.85 pF at a FB of 0.7V.).